

Operations Research

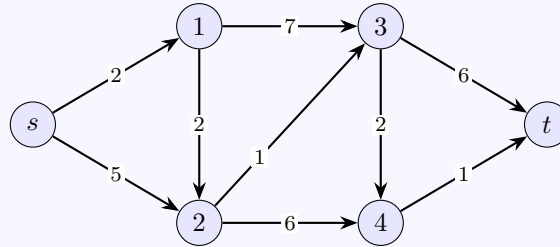
Problem Set G2: Graph Models (Bottlenecks, Bounds and Proofs)

Companion to Chapter 1, Sections 1.5–1.6

Scope. Harder instances of the four graph models, where the naive bound is *not* tight and capacities bind, plus the structural proofs behind the models. Solutions start on page 4.

Problem 1 (Fewest links versus least time)

A night convoy travels from s to t on the one-way network below; arc labels are travel times in minutes.

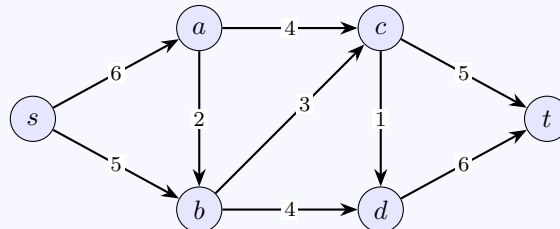


- Which routes use the fewest arcs? How long do they take?
- List all directed s – t paths with their times and find the fastest. How many arcs does it use? Comment on the difference with (a).
- Prove the fastest time is optimal by exhibiting labels d_v with $d_s = 0$ and $d_j \leq d_i + w_{ij}$ on every arc, as in Set G1. Which arcs satisfy the label inequality with equality, and what do they form?
- Road (2,3) closes for works. Without re-enumerating everything, find the new fastest route and time, and re-prove optimality with new labels.

Problem 2 (Maximum flow: where is the bottleneck?)

Data centre traffic flows from s to t through routers a, b, c, d . Capacities (Gbit/s):

$$u_{sa} = 6, \quad u_{sb} = 5, \quad u_{ab} = 2, \quad u_{ac} = 4, \quad u_{bc} = 3, \quad u_{bd} = 4, \quad u_{cd} = 1, \quad u_{ct} = 5, \quad u_{dt} = 6.$$



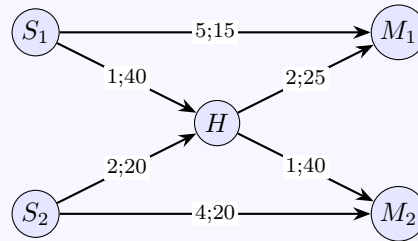
- Draw the network. Compute the capacity of the cuts $S = \{s\}$, $\{s, a\}$, $\{s, a, b\}$, $\{s, a, b, c\}$ and $\{s, a, b, c, d\}$.
- Find a flow whose value equals the smallest cut capacity in (a); verify conservation at a, b, c, d and every capacity. What is the maximum flow, and why?

- (c) The engineers may add 2 Gbit/s of capacity to *one* arc of their choice. Prove that no choice raises the maximum flow above 11. (Hint: which cuts have capacity 11, and do they share an arc?)
- (d) Exhibit one arc whose upgrade achieves 11, with the new flow.

Problem 3 (Minimum-cost flow with a binding capacity)

Two silos S_1 (40 t) and S_2 (20 t) supply two mills M_1 (35 t) and M_2 (25 t), directly or through a rail hub H . Arcs are labelled (cost per tonne; capacity):

Arc	Cost	Capacity
(S_1, H)	1	40
(S_2, H)	2	20
(S_1, M_1)	5	15
(H, M_1)	2	25
(H, M_2)	1	40
(S_2, M_2)	4	20



Name the flows $x_1 = x_{S_1H}$, $x_2 = x_{S_2H}$, $x_3 = x_{S_1M_1}$, $x_4 = x_{HM_1}$, $x_5 = x_{HM_2}$, $x_6 = x_{S_2M_2}$.

- Write the balance equations and the objective.
- Compute the “cheapest route per tonne” relaxation bound (drop supplies and capacities). Explain concretely why this bound cannot be attained here.
- Propose a plan, verify all constraints, and compute its cost.
- Use the balance equations to express x_1, x_2, x_3, x_6 in terms of x_4 and x_5 , and show that the cost equals $270 - 2x_4 - x_5$. Deduce the optimal cost and prove your plan in (c) is optimal. Which constraints are binding?
- The hub’s outbound line to M_1 is upgraded to capacity 30. What is the new optimal cost? How much is each extra tonne of that capacity worth?

Problem 4 (Assignment with more nurses than shifts)

Four nurses must cover three night shifts, one nurse per shift, each nurse at most one shift. Reluctance scores (lower is better):

	shift 1	shift 2	shift 3
N_1	3	5	9
N_2	2	6	2
N_3	7	1	8
N_4	5	4	3

- Write the model. Which constraints are equalities and which are inequalities, and why? Explain how adding a “dummy shift” with zero scores turns it into a standard assignment problem.
- Compute the column-minimum bound (sum over shifts of the cheapest nurse). Explain which constraints it relaxes and why it is not attained here.
- Find the optimal assignment(s) and score. Organise the enumeration so that you do not have to check all 24 possibilities.
- Show that if $u_i \leq 0$ for every nurse and $u_i + v_j \leq c_{ij}$ for every pair, then every feasible

assignment scores at least $\sum_i u_i + \sum_j v_j$. Find such numbers certifying your optimum in (c).

- (e) Nurse N_2 falls ill. What is the new optimum, and does the column-minimum bound now certify it?

Problem 5 (Structure of the models: proofs)

- (a) Let D be a finite digraph in which every node has at least one outgoing arc. Prove that D contains a directed cycle.
- (b) Let $x \in \{0, 1\}^A$ satisfy the shortest-path balance equations (+1 at s , -1 at t , 0 elsewhere). Prove that the arcs with $x_{ij} = 1$ contain a directed s - t path. (Hint: walk from s along selected arcs without reusing an arc; show you can always continue unless you are at t .)
- (c) Suppose all weights are strictly positive. Prove that every optimal solution of the shortest-path integer program is exactly the indicator vector of a single s - t path. (Hint: subtract the indicator of the path found in (b) and look at the cost of what remains.)
- (d) (Cut condition for flows with balances.) In the minimum-cost flow model, let S be any set of nodes and $b(S) = \sum_{i \in S} b_i$. Prove that feasibility requires $b(S) \leq u(\delta^+(S))$ and $-b(S) \leq u(\delta^-(S))$.
- (e) Apply (d) to the instance of Problem 3 with the capacity of (H, M_1) reduced from 25 to 15, with $S = \{S_1, S_2, H, M_2\}$. Conclude.

Solutions

Solution 1

- (a) Every s - t path has at least 3 arcs (no arc from s reaches t in two steps: $s \rightarrow 1 \rightarrow t$ and $s \rightarrow 2 \rightarrow t$ do not exist). Three-arc routes: $s24t$ ($5 + 6 + 1 = 12$) and $s13t$ ($2 + 7 + 6 = 15$).
- (b) Eight paths: $s1234t$: 8; $s123t$: 11; $s124t$: 11; $s134t$: 12; $s13t$: 15; $s234t$: 9; $s23t$: 12; $s24t$: 12. Fastest: $s \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow t$ in 8 minutes, using all five “short hops.” Fewest links (3) and least time (5 links) give different answers: the criterion is part of the model.
- (c) $d = (d_s, d_1, d_2, d_3, d_4, d_t) = (0, 2, 4, 5, 7, 8)$. Checks: $(s, 1)$: $2 \leq 2$; $(s, 2)$: $4 \leq 5$; $(1, 2)$: $4 \leq 4$; $(1, 3)$: $5 \leq 9$; $(2, 3)$: $5 \leq 5$; $(2, 4)$: $7 \leq 10$; $(3, 4)$: $7 \leq 7$; $(3, t)$: $8 \leq 11$; $(4, t)$: $8 \leq 8$. Telescoping along any path gives length $\geq d_t = 8$. Equality holds on $(s, 1), (1, 2), (2, 3), (3, 4), (4, t)$: exactly the arcs of the fastest path (in general, the tight arcs form a tree of shortest routes from s).
- (d) Remove the paths using $(2, 3)$: remaining $s124t$: 11, $s134t$: 12, $s13t$: 15, $s24t$: 12. New fastest: $s \rightarrow 1 \rightarrow 2 \rightarrow 4 \rightarrow t$, 11 minutes. Labels $(0, 2, 4, 9, 10, 11)$: $(1, 3)$: $9 \leq 9$; $(2, 4)$: $10 \leq 10$; $(3, 4)$: $10 \leq 11$; $(3, t)$: $11 \leq 15$; $(4, t)$: $11 \leq 11$; $(s, 2)$: $4 \leq 5$; the rest as before. Hence 11 is optimal.

Solution 2

- (a) $\{s\}$: $6 + 5 = 11$. $\{s, a\}$: $(s, b), (a, b), (a, c)$: $5 + 2 + 4 = 11$. $\{s, a, b\}$: $(a, c), (b, c), (b, d)$: $4 + 3 + 4 = 11$. $\{s, a, b, c\}$: $(b, d), (c, d), (c, t)$: $4 + 1 + 5 = 10$. $\{s, a, b, c, d\}$: $(c, t), (d, t)$: $5 + 6 = 11$.
- (b) $x_{sa} = 5, x_{sb} = 5, x_{ab} = 1, x_{ac} = 4, x_{bc} = 2, x_{bd} = 4, x_{cd} = 1, x_{ct} = 5, x_{dt} = 5$. Conservation: a : in 5, out $1 + 4$; b : in $5 + 1$, out $2 + 4$; c : in $4 + 2$, out $1 + 5$; d : in $4 + 1$, out 5. All capacities hold. Value 10, equal to the capacity of $\{s, a, b, c\}$, so 10 is the maximum flow (cut bound plus matching).
- (c) After upgrading a single arc by 2, the four cuts of capacity 11 in (a) have arc sets $\{sa, sb\}$, $\{sb, ab, ac\}$, $\{ac, bc, bd\}$, $\{ct, dt\}$. No arc lies in all four (indeed none lies in both $\{sa, sb\}$ and $\{ct, dt\}$), so at least one of them keeps capacity 11, and the cut bound gives max flow ≤ 11 .
- (d) Upgrade (b, d) to 6. Push one more unit along $s \rightarrow a \rightarrow b \rightarrow d \rightarrow t$: $x_{sa} = 6, x_{ab} = 2, x_{bd} = 5, x_{dt} = 6$, rest unchanged. Check b : in $5 + 2 = 7$, out $2 + 5 = 7$; d : in $5 + 1 = 6$, out $6 \leq 6$. Value 11, which is maximal by (c). (Upgrading (c, t) or (c, d) also works.)

Solution 3

- (a) S_1 : $x_1 + x_3 = 40$; S_2 : $x_2 + x_6 = 20$; H : $x_4 + x_5 - x_1 - x_2 = 0$; M_1 : $-x_3 - x_4 = -35$; M_2 : $-x_5 - x_6 = -25$. Objective min $x_1 + 2x_2 + 5x_3 + 2x_4 + x_5 + 4x_6$, with $0 \leq x_1 \leq 40, x_2 \leq 20, x_3 \leq 15, x_4 \leq 25, x_5 \leq 40, x_6 \leq 20$.
- (b) Cheapest route to M_1 : S_1HM_1 at 3 (others 4, 5); to M_2 : S_1HM_2 at 2 (others 3, 4). Bound $35 \cdot 3 + 25 \cdot 2 = 155$. Unattainable: all 35 tonnes for M_1 would have to use (H, M_1) , whose capacity is 25; at least 10 tonnes must take the direct road at cost 5. Also S_1 would have to ship $60 > 40$ tonnes.
- (c) $x_4 = 25, x_3 = 10, x_1 = 30, x_5 = 25, x_2 = 20, x_6 = 0$. Balances: S_1 : $30 + 10$; S_2 :

$20 + 0$; H : $25 + 25 = 30 + 20$; M_1 : $10 + 25$; M_2 : $25 + 0$. Capacities fine ($x_4 = 25 \leq 25$, $x_2 = 20 \leq 20$). Cost $30 + 40 + 50 + 50 + 25 + 0 = 195$.

- (d) From M_1 : $x_3 = 35 - x_4$; from S_1 : $x_1 = 5 + x_4$; from M_2 : $x_6 = 25 - x_5$; from S_2 : $x_2 = x_5 - 5$ (the hub equation is then automatically satisfied). Substituting,

$$(5 + x_4) + 2(x_5 - 5) + 5(35 - x_4) + 2x_4 + x_5 + 4(25 - x_5) = 270 - 2x_4 - x_5.$$

Constraints: $x_4 \leq 25$ (capacity of (H, M_1)) and $x_5 \leq 25$ (from $x_6 \geq 0$; also $x_2 \leq 20$), so cost $\geq 270 - 50 - 25 = 195$. The plan in (c) attains it: optimal. Binding: capacity of (H, M_1) , the bound $x_6 \geq 0$ (equivalently $x_5 \leq 25$), and $x_2 \leq 20$.

- (e) Now $x_4 \leq 30$ is allowed (and $x_1 = 5 + x_4 = 35 \leq 40$, $x_3 = 5 \geq 0$ remain fine), so cost $= 270 - 60 - 25 = 185$. Each extra tonne of capacity on (H, M_1) saves 2 (it moves one tonne from the 5-cost direct road to the 3-cost hub route), up to $x_4 = 35$ where x_3 hits 0.

Solution 4

- (a) $\min \sum c_{ij}x_{ij}$ s.t. $\sum_i x_{ij} = 1$ for each shift j (every shift must be covered), $\sum_j x_{ij} \leq 1$ for each nurse i (a nurse may rest), $x_{ij} \in \{0, 1\}$. Adding a dummy shift 0 with $c_{i0} = 0$ and requiring $\sum_j x_{ij} = 1$ makes it a 4×4 assignment problem; the nurse assigned to shift 0 is the one who rests.
- (b) Column minima: shift 1: 2 (N_2), shift 2: 1 (N_3), shift 3: 2 (N_2); bound 5. It relaxes the nurse constraints, and it is not attained because N_2 would have to take two shifts.
- (c) Shift 2 is cheapest with N_3 (1); any other choice costs ≥ 4 . *Case $N_3 \rightarrow 2$* : choose distinct nurses from $\{N_1, N_2, N_4\}$ for shifts 1, 3: (N_1, N_2) : $3+2=5$; (N_2, N_4) : $2+3=5$; (N_1, N_4) : 6; (N_4, N_2) : 7; (N_2, N_1) : 11; (N_4, N_1) : 14. Best total $1 + 5 = 6$. *Case $N_3 \not\rightarrow 2$* : shift 2 costs ≥ 4 and shifts 1, 3 cost at least $2+2$ but not both by N_2 , so at least $2+3=5$; total ≥ 9 . Optimum 6, attained by $N_1 \rightarrow 1, N_3 \rightarrow 2, N_2 \rightarrow 3$ and by $N_2 \rightarrow 1, N_3 \rightarrow 2, N_4 \rightarrow 3$.
- (d) $\sum c_{ij}x_{ij} \geq \sum_i u_i \sum_j x_{ij} + \sum_j v_j \sum_i x_{ij} \geq \sum_i u_i + \sum_j v_j$, because $\sum_j x_{ij} \leq 1$ and $u_i \leq 0$ give $u_i \sum_j x_{ij} \geq u_i$, and $\sum_i x_{ij} = 1$. Take $v = (3, 1, 3)$ and $u = (0, -1, 0, 0)$: row N_2 needs $2 \geq 3 - 1$, $6 \geq 0$, $2 \geq 2$; the other rows need $c_{ij} \geq v_j$, which holds. Bound $= 7 - 1 = 6$: optimal.
- (e) Without N_2 : shift 2 $\rightarrow N_3$ (1), shift 1 $\rightarrow N_1$ (3), shift 3 $\rightarrow N_4$ (3): total 7. Column minima are now 3, 1, 3 from three *different* nurses, so the bound 7 is attained and certifies optimality directly.

Solution 5

- (a) Start anywhere and keep following an outgoing arc. The walk never gets stuck, but there are finitely many nodes, so some node repeats; the portion of the walk between two visits of that node is a directed cycle.
- (b) Walk from s along selected arcs, never reusing an arc. At s : out = in + 1; after the k -th arrival we have used k incoming and k outgoing selected arcs (the first departure counts), so an unused outgoing arc remains. At a node $v \notin \{s, t\}$: out = in, and after the k -th arrival we have used k incoming and $k - 1$ outgoing, so we can leave. The walk uses each arc at most once, hence stops, and it can only stop at t . Deleting the portions between repeated visits of the same node turns the walk into a path.

- (c) Let x be optimal and P the path from (b), with indicator $x^P \leq x$. Then $y = x - x^P \in \{0, 1\}^A$ has balance 0 at every node, so y is feasible for the “send nothing” version and $w^\top y \geq 0$, with $w^\top y > 0$ unless $y = 0$ (weights positive). Since x^P is feasible for the shortest-path program and $w^\top x = w^\top x^P + w^\top y$, optimality of x forces $w^\top y = 0$, so $y = 0$ and $x = x^P$.
- (d) Sum the balance equations over $i \in S$: internal arcs cancel and we get $x(\delta^+(S)) - x(\delta^-(S)) = b(S)$. Hence $b(S) \leq x(\delta^+(S)) \leq u(\delta^+(S))$ and $-b(S) \leq x(\delta^-(S)) \leq u(\delta^-(S))$.
- (e) $b(S) = 40 + 20 + 0 - 25 = 35$; $\delta^+(S) = \{(S_1, M_1), (H, M_1)\}$ with capacity $15 + 15 = 30 < 35$. The condition fails, so the reduced instance is infeasible: M_1 cannot receive 35 tonnes through 30 tonnes of inbound capacity.